

Skills Summary

Holes and Vertical Asymptotes

Use this reference while completing your worksheet or when studying.

Skill 1 — Factor top and bottom, then cancel

Rule: a rational expression can only be simplified by cancelling a *whole factor* that appears in both the numerator and the denominator — never a single term out of a sum. Factor completely first, cancel the shared factors, and keep the original denominator's zeros in mind: cancelling changes the formula, not the domain.

Example: Simplify $\frac{x^2 - 16}{x^2 - 2x - 8}$.

Step 1. Nothing cancels while the parts are sums, so factor the top and the bottom completely before looking for anything to divide out.

Step 2. $\frac{x^2 - 16}{x^2 - 2x - 8} = \frac{(x - 4)(x + 4)}{(x - 4)(x + 2)}$, and the factor $(x - 4)$ is shared.
 $\Rightarrow \frac{x + 4}{x + 2}$

Try it: Simplify $\frac{x^2 - 25}{x^2 + x - 20}$.

$\frac{4 - x}{5 - x}$ check

Skill 2 — Hole or vertical asymptote?

Test, applied to each zero of the ORIGINAL denominator:

- its factor **cancels** $\Rightarrow$ **hole**: one point missing from an otherwise ordinary curve;
- its factor **survives** $\Rightarrow$ **vertical asymptote**: the values run off without bound.

Both kinds stay out of the domain. If every denominator factor cancels, the graph has no vertical asymptote at all.

Example: For $f(x) = \frac{x^2 + 3x - 4}{x^2 - 6x + 5}$, which denominator zero gives a hole and which gives a vertical asymptote?

Step 1. The decision is made by the factored form, so factor both parts and see which factor divides out and which one is left downstairs.

Step 2. $\frac{(x - 1)(x + 4)}{(x - 1)(x - 5)}$: the factor $(x - 1)$ cancels, so $x = 1$ is a hole, while $(x - 5)$ survives, so $x = 5$ is a vertical asymptote.

$\Rightarrow f(x) = \frac{x + 4}{x - 5}$ (hole at $x = 1$, asymptote at $x = 5$)

Try it: For $g(x) = \frac{x^2 - 2x - 8}{x^2 - 16}$, simplify, then name the hole and the asymptote.

check: $\frac{x+2}{x-4}$, hole at $x = 4$, asymptote at $x = -4$

Skill 3 — Finding the height of a hole

Rule: the hole sits at the x -value of the cancelled factor, and its height is the value of the *simplified* expression there. Substituting into the original gives $\frac{0}{0}$, which is not 0 and not an answer.

Example: Find the coordinates of the hole in $f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$.

Step 1. Both parts vanish at the same x , which is the signature of a hole, so cancel the shared factor and read the height off what is left.

Step 2. $\frac{(x-2)(x+2)}{(x-2)(x-3)} = \frac{x+2}{x-3}$; the cancelled factor puts the hole at $x = 2$, and there

$$\frac{2+2}{2-3} = \frac{4}{-1}.$$

$\Rightarrow$ hole at $(2, -4)$

Try it: Find the coordinates of the hole in $g(x) = \frac{x^2 - 1}{x^2 + 2x - 3}$.

check: $(\frac{x-1}{x+3})$, hole at $x = 1$